Incident Report: Storm Power Failure in the Nursery and Reptile Incubation Wing

Huai Mae Kham Wildlife Rescue Centre, Chiang Rai · Ref. IR-2026-07 · Version 1.1 · Issued 2 September 2026

1. Summary

On the night of 14–15 August 2026 a thunderstorm cut mains power to the whole Huai Mae Kham Wildlife Rescue Centre for 5 hours and 40 minutes, from 23:10 on 14 August to 04:50 on 15 August. The standby diesel generator (asset GEN-02, 60 kVA) started correctly, but the automatic transfer switch did not connect it to the East Wing, which houses the Mammal Nursery (Rooms E1 and E2) and the Reptile Incubation Room (Room E4). The East Wing therefore stayed without power for 2 hours and 25 minutes, until staff switched it over by hand at 01:35.

During that time the temperature in the Reptile Incubation Room fell from 29.5 °C to 24.1 °C, and the Mammal Nursery fell from 30 °C to 25.8 °C. No animal died during the outage. Of the 41 forest tortoise eggs in incubation, 6 later failed to hatch; the reptile team believes the cold period contributed to at least 4 of these losses. Two infant slow lorises and one infant binturong were moved to a heated transport cabinet and recovered fully.

This report was prepared by the Incident Review Group, which met on 21 and 28 August 2026. Version 1.0 was circulated internally on 29 August; version 1.1 corrects the duration of the East Wing outage, which was first reported as 2 hours and 10 minutes.

2. Timeline of the night

- **22:40** – The regional weather service issues a severe thunderstorm warning for Mueang Chiang Rai and Mae Lao districts.
- **23:10** – Mains power fails across the site. GEN-02 starts within 12 seconds and powers the West Wing, the clinic and the kitchen.
- **23:25** – Night keeper Thanakrit Moonsri notices that the East Wing corridor lights are still off and that the incubator alarm panel in Room E4 is running on its internal battery.
- **23:40** – The night keeper phones the on-call facilities technician, who does not answer. The on-call list on the notice board still showed the June rota.
- **00:05** – The night keeper reaches the duty veterinarian, Dr. Orawan Boonyasit, who arrives on site at 00:30.
- **00:45** – On Dr. Orawan's instruction, the two slow loris infants and the binturong infant are moved from Room E1 into the heated transport cabinet in the clinic.
- **01:20** – Facilities Supervisor Chaiyaporn Phanphet arrives after being called by Dr. Orawan.
- **01:35** – Mr. Chaiyaporn switches the East Wing to generator power using the manual bypass handle on the transfer switch. Heating returns in Rooms E1, E2 and E4.
- **03:00** – Room E4 is back to 29.0 °C. The incubators are checked one by one.
- **04:50** – Mains power returns. The transfer switch is left in manual mode until inspection.

Fictional document created for testing a question-answering system. All names, figures and organisations are invented.

3. Effect on animals

At the time of the outage the East Wing held 9 animals in the Mammal Nursery and 41 forest tortoise eggs in three incubators in Room E4. The eggs came from two confiscated clutches received on 22 July 2026 and were due to hatch between 2 and 20 September.

- **Mammal Nursery:** all 9 animals survived. The two slow lorises and the binturong, which are under 6 weeks old and cannot regulate their body temperature, were moved first. The six older animals stayed in their rooms with extra bedding.
- **Reptile Incubation Room:** by 14 September, 35 eggs had hatched and 6 had failed. Hatchlings are being monitored for shell deformities until 30 November 2026.
- **West Wing and outdoor enclosures:** not affected. One fallen branch damaged the fence of the macaque pre-release enclosure, which was repaired on 16 August.

Note. The incubators in Room E4 have their own alarm batteries but no heating backup. The alarm panel showed a low-temperature warning from 23:58, but the alarm only sounds locally in Room E4 and is not linked to the night keeper's radio.

4. Cause of the failure

An inspection by the generator service contractor on 18 August found that the control relay in the East Wing section of the automatic transfer switch was corroded and had stuck in the open position. The relay was last checked during the annual service on 9 November 2025. Under the centre's maintenance plan, the transfer switch should also have been load-tested every three months; the last recorded load test was on 3 February 2026, so the May 2026 test was missed.

The review also found three contributing factors:

- The on-call facilities rota on the notice board had not been updated since June, so the night keeper called a technician who had left the centre on 31 July.
- Night keepers had not been trained to use the manual bypass handle. The handle is behind a locked panel in the plant room and only facilities staff hold the key.
- The written power-failure procedure, dated 2023, did not mention the East Wing as a separate circuit.

5. Revised power-failure procedure

The procedure below replaces the 2023 version with immediate effect. A laminated copy is posted in the night keeper's office and in Rooms E1, E2 and E4. All night keepers and duty veterinarians must confirm in writing that they have read it.

1. Within 10 minutes of any power cut, the night keeper checks the indicator lamps on the transfer switch display in the plant room and walks to the East Wing to confirm that lights and heaters are on.
2. If the East Wing has no power, the night keeper calls the on-call facilities technician from the current rota in the shared roster app, not from the notice board.
3. If the technician does not answer within 15 minutes, the night keeper calls the Facilities Supervisor directly and then the duty veterinarian.

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4. If the temperature in Room E4 or the Mammal Nursery falls below 27 °C, the night keeper moves infants under 6 weeks old to the heated transport cabinet in the clinic, without waiting for the duty veterinarian.
5. Only facilities staff, or a night keeper who has passed the bypass training, may operate the manual bypass handle. The key is now kept in the sealed key box in the night keeper's office.
6. The night keeper records each step and its time in the incident log and informs the Centre Director by message before 07:00.

Note. Incubators must not be opened during a power cut, even to check eggs, unless the duty veterinarian or the Reptile Unit Lead gives permission. Opening a cold incubator lowers the temperature further.

6. Responsibilities

The review group confirmed who is responsible for each part of power resilience at the centre:

- **Facilities Supervisor (Chaiyaporn Phanphet):** quarterly load tests of GEN-02 and the transfer switch, the annual generator service contract, the on-call facilities rota in the roster app, and bypass training for night keepers.
- **Senior Veterinarian (Dr. Orawan Boonyasit):** decisions on moving or treating animals during an emergency, and the animal welfare section of every incident report.
- **Reptile Unit Lead (Sirilak Pansuwan):** incubator settings, egg checks and hatchling monitoring.
- **Centre Director (Jiraporn Kaewlamai):** approval of this report, notification of the foundation board within 72 hours of any incident that affects animal welfare, and the budget for corrective actions.

7. Corrective actions

- Replace the corroded relay and fit a new relay rated for humid conditions. Owner: Facilities Supervisor. Completed on 19 August 2026 at a cost of 8,400 baht.
- Install a battery-backed heater (minimum 4 hours) in each incubator in Room E4. Owner: Reptile Unit Lead. Due 15 October 2026; budget approved at 36,000 baht for three units.
- Link the Room E4 and Mammal Nursery temperature alarms to the night keeper's radio. Owner: Facilities Supervisor. Due 30 September 2026.
- Train all five night keepers on the manual bypass handle, with a practical test. Owner: Facilities Supervisor. Two sessions are scheduled for 10 and 17 September 2026.
- Run an unannounced power-failure drill on a night shift. Owner: Centre Director. To be held before 31 December 2026.

The Centre Director informed the foundation board of the incident on 16 August 2026. Progress on the actions above will be reviewed at the monthly safety meeting on 7 October 2026, and this report will then be closed or updated as version 1.2.

Prepared by the Incident Review Group: Dr. Orawan Boonyasit, Chaiyaporn Phanphet, Sirilak Pansuwan and Thanakrit Moonsri. Approved by Jiraporn Kaewlamai, Centre Director, on 2 September 2026.

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